



H2S

BHARATIYA ANTARIKSH HACKATHON 2026



Team Name : Bugs2Bucks

Team Leader Name : Karan A Sharma

Problem Statement :Optimizing Urban Heat Mitigation and cooling strategies via Artificial Intelligence and Machine Learning (AIML)

Meet Team **Bugs2Bucks**

A passionate team building AI-powered solutions for urban heat mitigation



Karan A Sharma

Team Lead & Full-Stack Developer



Product Strategy



AI Integration



System Architecture



Pratik K Koradur

ML & Data Engineering



AI/ML Models



Data Processing



Geospatial Analytics



Kumar Satyam

Frontend & Visualization



UI/UX Design



Dashboard Development



Data Visualization



Dayananda Sagar University, Bengaluru



Building AI for **Smarter & Cooler Cities**



Innovation & Competitive Advantage



1. DIFFERENTIATION

Existing tools (Google Environmental Insights Explorer, Climate-ADAPT, ESRI ArcGIS Urban Heat) focus on **heat mapping and hotspot visualization**.

THERMOS AI goes further by **explaining** heat drivers, **simulating** cooling interventions, and **recommending** optimal mitigation strategies with predicted impact.

01



2. PROBLEM-SOLVING APPROACH

Integrates satellite, weather, land-use, and urban infrastructure data with AI/ML models to **identify heat hotspots, analyze causes, and recommend effective cooling interventions**.

02



3. USP

THERMOS AI moves beyond heat hotspot detection by combining **satellite intelligence, AI-driven analysis, and what-if simulations** into a single decision-support platform.

Enables **data-driven** urban cooling decisions before implementation.

03



Predict. Simulate. Optimize. Cool Cities Smarter.



Key Features of THERMOS AI

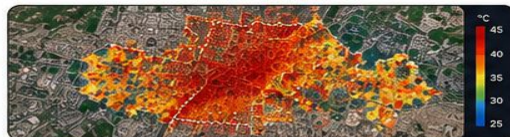
An AI-powered platform for data-driven urban heat mitigation

01



Heat Hotspot Detection

Identify urban heat stress zones using satellite and weather data.



02



Heat Driver Analysis

Analyze factors such as vegetation, land use, building density, and atmospheric conditions.



03



AI-Powered Predictions

Predict urban heat patterns and future risk zones using AI/ML models.



04



Cooling Scenario Simulator

Test interventions like tree cover, cool roofs, green corridors, and water bodies.



05



Decision Support Dashboard

Visualize hotspots, insights, recommendations, and impact metrics in one platform.



06



Intervention Optimization

Recommend the most effective cooling strategy with estimated temperature reduction.

Recommended Strategy		Est. Temp Reduction
Increase Tree Cover		-2.8°C
Cool Roof Implementation		-2.1°C
Green Corridors		-1.9°C
Water Bodies Restoration		-1.6°C



Detect



Analyze



Predict



Simulate



Optimize



Mitigate

THERMOS AI

PROCESS FLOW DIAGRAM

1 DATA SOURCES

Multi-source data collection from satellites, sensors, and open datasets



Raw Data Ingestion

2 DATA PROCESSING

Clean, normalize and integrate data for accurate analysis



Processed, High-quality Datasets

3 AI ANALYTICS ENGINE

AI/ML models for urban heat intelligence

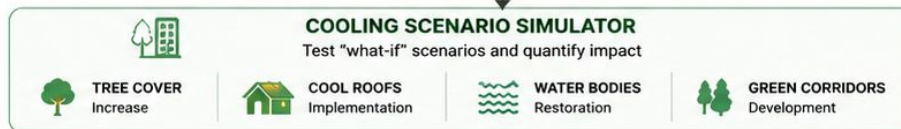


AI-powered Insights & Predictions



4 SCENARIO SIMULATOR

Simulate and evaluate cooling interventions



Scenario Results & Impact Estimates

5 RECOMMENDATION ENGINE

AI-powered optimization and prioritization



Optimal, Cost-effective Cooling Strategies

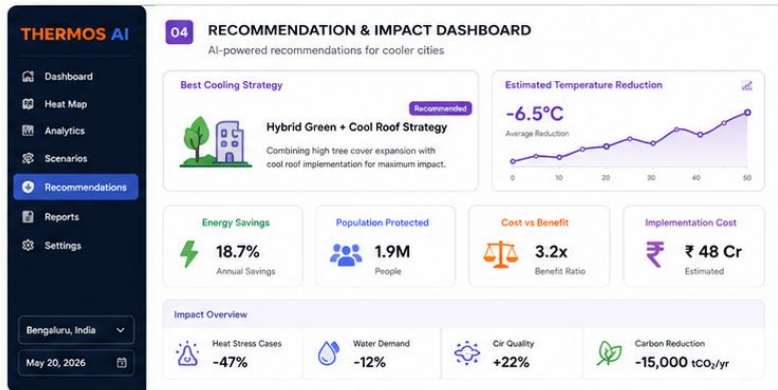
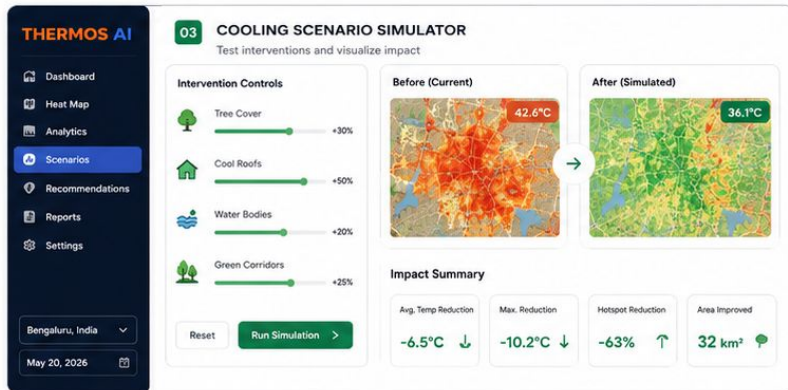
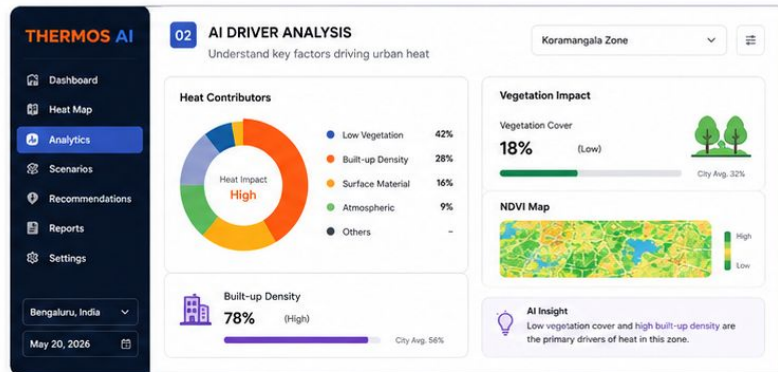
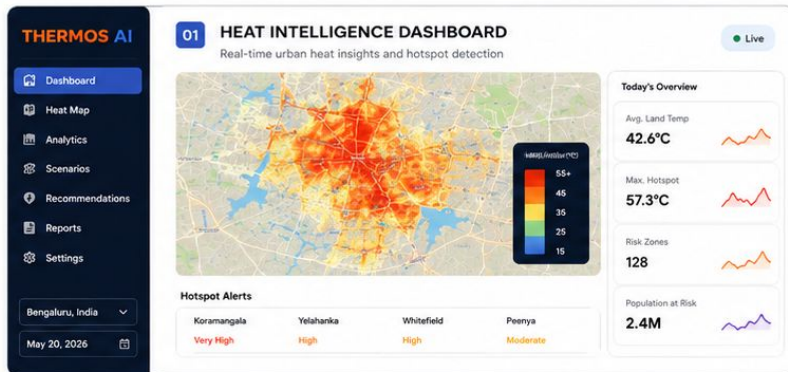
6 OUTPUT & IMPACT

Actionable insights for decision makers



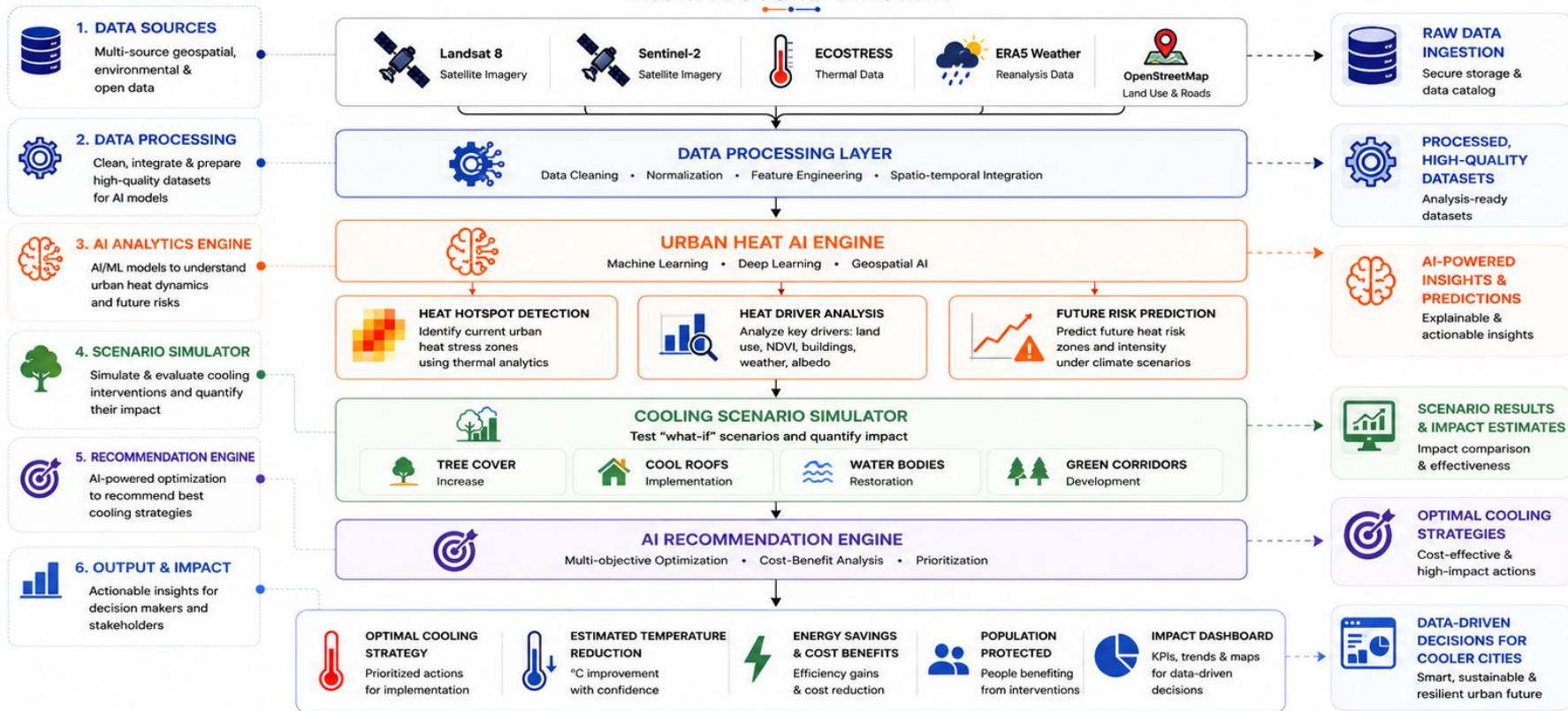
Data-driven Decisions for Cooler Cities

Wireframes/Mock diagrams of the proposed solution (optional)



THERMOS AI

ARCHITECTURE DIAGRAM



Technologies to be used in the solution:

1. DATA SOURCES



- Landsat 8
- Sentinel-2
- ECOSTRESS
- ERA5 Weather
- OpenStreetMap

2. DATA PROCESSING



- Google Earth Engine
- Python (PyQGIS, Rasterio)
- GDAL
- Pandas, NumPy
- PostGIS

3. AI & ANALYTICS ENGINE



- Python
- PyTorch / TensorFlow
- Scikit-learn
- XGBoost
- SHAP (Explainable AI)

4. SCENARIO SIMULATOR



Tree Cover

- Prioritization Modeling
- i-Tree / UFORE
- GeoPandas



Cool Roofs

- Albedo Modeling
- ENVI-met (optional)
- Python



Water Bodies

- Surface Water Mapping
- JRC / MODIS
- Sentinel-2



Green Corridors

- Network Analysis
- Graph Tools
- GeoNetworkX

5. RECOMMENDATION ENGINE



- Multi-objective Optimization
- Optuna / PyGMO
- Cost-Benefit Analysis
- Prioritization Ranking

6. PLATFORM & APPLICATION LAYER



Web Application

- Next.js (React)
- Tailwind CSS
- TypeScript



Backend Services

- FastAPI (Python)
- RESTful APIs
- Uvicorn



Databases

- PostgreSQL + PostGIS
- MongoDB
- Redis (Caching)



Cloud & Deployment

- AWS / GCP
- Docker
- Kubernetes



Authentication & Security

- Keycloak / OAuth2
- Role-based Access
- Data Encryption

7. VISUALIZATION & MAPS



- Mapbox / Leaflet.js
- Deck.gl / Kepler.gl
- Chart.js / D3.js



- Interactive Dashboards
- Real-time Monitoring
- Impact Analytics

8. TOOLS & DEVOPS



- Git & GitHub
- CI/CD (GitHub Actions)
- Monitoring (Prometheus + Grafana)
- Logging (ELK Stack)

Estimated implementation cost (optional):

Component	Description	Estimated Cost (INR)
 Cloud Hosting	Vercel / Supabase / AWS Free Tier	₹3,000
 Domain & Deployment	Domain, SSL, CI/CD setup	₹2,000
 AI/ML Processing	Model training, inference & APIs	₹5,000
 Dashboard & Visualization	Maps, charts, 3D city visualization	₹3,000
 Testing & Miscellaneous	Testing, logging, monitoring	₹2,000
 Total Estimated Prototype Cost		₹15,000



COST SUMMARY

Total Estimated Cost (MVP)

₹15,000



Affordable, scalable and built using open-source technologies



COST OPTIMIZATION

- 100% Open-source datasets (Landsat, Sentinel-2, ERA5, OSM)
- Open-source AI frameworks (Python, PyTorch, Scikit-learn)
- Student developer team
- Cloud free tiers and credits
- No proprietary satellite data required



Estimated MVP implementation cost: ₹10,000 – ₹15,000
using open-source technologies and publicly available datasets.



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THANK YOU